TITANIC SURVIVAL ANALYSIS

EXPLORATORY DATA ANALYSIS

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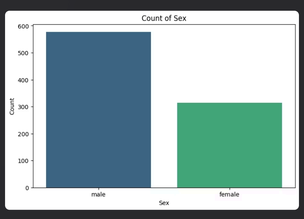
Made with GAMMA

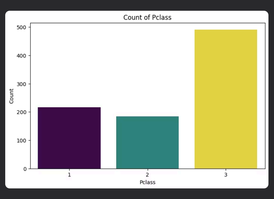
CONTENTS

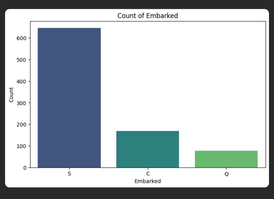
Dataset Overview

- 891 passengers (train.csv)
- 12 columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- Target: Survived (0/1) – 38.4% positive class
- Memory: ~83.7 KB

Categorical Distributions







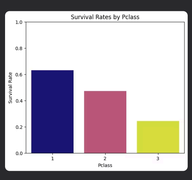
- Sex: ~65% male (577), ~35% female (314)
- Pclass: majority 3rd class
- Embarked: S (most common), C, Q

Missing Values & Cleaning

- Age: 177 missing → imputed with median (28–29)
- Embarked: 2 missing → imputed with mode ('S')
- Cabin: 687 missing (~77%) → column dropped
- Result: 0 missing values after cleaning

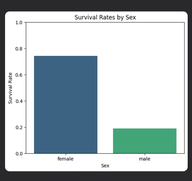
Overall Survival Rate

Survival rate: 38.38% (342/891)



Survival by Passenger Class

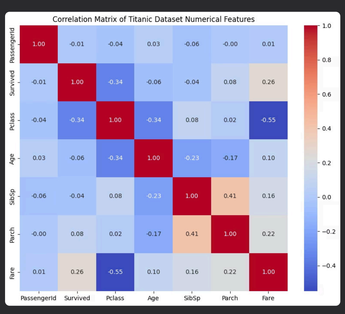
- 1st class: 63.0%
- 2nd class: 47.3%
- 3rd class: 24.2%



Survival by Gender

- Female: 74.2%
- Male: 18.9%

Correlation Heatmap



- Cherbourg (C) → highest survival rate
- Higher Fare → strongly associated with survival
- Strongest negative: Pclass ,<=> Survived & Pclass <=> Fare
- Sex (encoded) and Pclass most predictive of Survived

Key Insight

- Gender & socio-economic class (Pclass/Fare) were decisive
- "Women and children first" policy clearly visible
- Most passengers traveled alone → family features may help modeling

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